Plan

- Linear Transformation
- Kernel and Range
- The Rank-Nullity Theorem
- Isomorphism
- The Matrix of a Linear Transformation

Linear Transformations

Definition

Let V and W be vector spaces, both over $\mathbb{F}$, and $T : V \to W$. Then T is called a linear transformation from V to W if for all $\mathbf{u}, \mathbf{v} \in V$ and $a \in \mathbb{F}$

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \text{ and } T(a\mathbf{v}) = aT(\mathbf{v}).$$

(T respects the additions and scalar multiplications!)

Equivalently, $T(a\mathbf{u} + \mathbf{v}) = aT(\mathbf{u}) + T(\mathbf{v})$.

Example

For any V, W over $\mathbb{F}$, $T_0 : V \to W$, defined by $T_0(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in V$. The map T_0 is called the zero transformation.

Example

For any V over $\mathbb{F}$, $I : V \to V$, defined by $I(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in V$. The map I is called the identity transformation of V.

Example

Are the following linear transformations?

- $T : \mathbb{R}^2 \to \mathbb{R}^2$, *defined by* $T([x, y]^t) = [2x, x + y]^t$.
- $T : \mathbb{R}^2 \to \mathbb{R}$, *defined by* $T([x, y]^t) = xy$.
- $T : \mathbb{R}[x] \to \mathbb{R}[x]$ *given by* $T(p(x)) = \frac{d}{dx}p(x)$.
- $T : \mathbb{R}[x] \to \mathbb{R}$ *given by* $T(p(x)) = p(3)$.
- $T : M_2(\mathbb{R}) \to \mathbb{R}$ *given by* $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a + d$.
- $T : M_2(\mathbb{R}) \to \mathbb{R}$ *given by* $T(A) = \det(A)$.
- A *is a fixed matrix in* $M_2(\mathbb{R})$, $T : M_2(\mathbb{R}) \to M_2(\mathbb{R})$ *given by* $T(M) = AM$.

Result

Let $T : V \to W$ be a linear transformation. Then

1. $T(\mathbf{0}) = \mathbf{0}$;
2. $T(-\mathbf{v}) = -T(\mathbf{v})$ *for all* $\mathbf{v} \in V$; *and*
3. $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$ *for all* $\mathbf{u}, \mathbf{v} \in V$.

Exercise

Is the map $T : \mathbb{R} \to \mathbb{R}$, *defined by* $T(x) = x + 1$ *for all* $x \in \mathbb{R}$ *a linear transformation?*

Exercise

Is there a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}_2[x]$ *such that* $T[1,0]^t = 2 - 3x$, $T[0,1]^t = 1 - x^2$ *and* $T[2,2]^t = 5 - 6x - 2x^2$?

Example

Let A be an $m \times n$ matrix. Define $T_A : \mathbb{R}^n \to \mathbb{R}^m$ such that $T_A(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. Then T_A is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$. We say T_A is induced by A.

Example

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation. Let $T(\mathbf{e}_1) = [a, b]^t$, $T(\mathbf{e}_2) = [c, d]^t$. Then

$$T([x, y]) = T(x\mathbf{e}_1 + y\mathbf{e}_2) = x \begin{bmatrix} a \\ b \end{bmatrix} + y \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Exercise

Let $T : \mathbb{F}^n \to \mathbb{F}^m$ be a linear transformation. Is $T = T_A$ for some $m \times n$ matrix A?

Result

Let $T : V \to W$ *be a linear transformation,* $\mathbf{v}_1, \ldots, \mathbf{v}_k \in V$ *and* $c_1, \ldots, c_k \in \mathbb{F}$. *Then*

$$T(c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k) = c_1 T(\mathbf{v}_1) + \cdots + c_k T(\mathbf{v}_k).$$

Exercise

Suppose $T : \mathbb{R}^2 \to \mathbb{R}_2[x]$ *is a linear transformation such that* $T[1,0]^t = 2 - 3x + x^2$ *and* $T[0,1]^t = 1 - x^2$.
Find $T[2,3]^t$ *and* $T[a,b]^t$.

Result

Let V, W *be vector spaces over* $\mathbb{F}$, *and* $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ *be a basis of* V. *Let* $\mathbf{w}_1, \ldots, \mathbf{w}_n$ *be arbitrarily chosen in* W. *Then there is a unique linear transformation* $T : V \longrightarrow W$ *such that* $T(\mathbf{v}_i) = \mathbf{w}_i$.

Kernel and Range

Definition

Let $T : V \to W$ be a linear transformation. Then the kernel (or the null space) of T is defined as

$$\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}$$

The range of $T$is defined as

$$\begin{aligned} \text{range}(T) &= \{T(\mathbf{v}) : \mathbf{v} \in V\} \\ &= \{\mathbf{w} \in W : \mathbf{w} = T(\mathbf{v}) \text{ for some } \mathbf{v} \in V\} \end{aligned}$$

- $\ker(T)$ is a subspace of V.
- $\text{range}(T)$ is a subspace of W.

Result

A linear transformation $T : V \to W$ is one-one iff $\ker(T) = \{\mathbf{0}\}$.

- $T : V \to W$ is onto iff $\text{range}(T) = W$.

Example

Consider $T_A : \mathbb{R}^n \to \mathbb{R}^m$, *where A is an $m \times n$ matrix, i.e.,* $T_A(\mathbf{x}) = A\mathbf{x}$. *Then*

$$ker(T) = null(A) \text{ and } range(T) = col(A).$$

Exercise

Let A be an $m \times n$ matrix. When is $T_A : \mathbb{F}^n \to \mathbb{F}^m$ one-one? Onto? If $n > m$, can T_A be one-one? If $n < m$, can T_A be onto?

Result

Let $T : V \to W$ *be a linear transformation, and* $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\} \subseteq V$

- *If S spans V, then $T(\mathbf{v}_1), \ldots, T(\mathbf{v}_k)$ span $range(T)$.*
- *If $T(\mathbf{v}_1), \ldots, T(\mathbf{v}_k)$ are linearly independent, then S is linearly independent.*

Definition

Let $T : V \to W$ be a linear transformation. We define

- $\text{rank}(T) =$ dimension of range(T); and
- $\text{nullity}(T) =$ dimension of $\ker(T)$.

Example

Let $D : \mathbb{R}_3[x] \to \mathbb{R}_2[x]$ be defined by $D(p(x)) = \frac{d}{dx}p(x)$. Then $rank(D) = 3$ and $nullity(D) = 1$.

Exercise

If $T = T_A$ for some matrix A, then what are $rank(T)$ and $nullity(T)$?

Result (**The Rank-Nullity Theorem**)

Let $T : V \to W$ be a linear transformation from a finite dimensional vector space V into a vector space W. Then

$$rank(T) + nullity(T) = dim(V).$$

Take Home

Exercise

Let dim$(V) =$ dim(W). Then a linear transformation $T : V \to W$ is one-one iff T is onto.

Exercise

Let $T : V \to W$ be a one-one linear transformation. If $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is a linearly independent set in V then $T(S) = \{T(\mathbf{v}_1), \ldots, T(\mathbf{v}_k)\}$ is a linearly independent set in W.

Exercise

Let dim$(V) =$ dim(W). Then a one-one linear transformation $T : V \to W$ maps a basis for V onto a basis for W.

Linear Isomorphisms

- A linear transformation $T : V \to W$ is called an isomorphism if it is one-one and onto, i.e, if $\ker(T) = \{\mathbf{0}\}$ and $\mathrm{range}(T) = W$.
- If there is an isomorphism from V to W, then we say that V is isomorphic to W, and we write $V \cong W$.

Example

The vector spaces $\mathbb{R}^3$ and $\mathbb{R}_2[x]$ are isomorphic.

Exercise

Let $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be an ordered basis of V. Consider $T : V \to \mathbb{F}^n$ defined by $T(\mathbf{v}) = [\mathbf{v}]_B$.
Is it a linear transformation? Is it an isomorphism?

Exercise

Suppose $T : V \to W$ is an isomorphism and $\mathbf{v}_1, \ldots, \mathbf{v}_n$ form a basis for V. What can you say about $T(\mathbf{v}_1), \ldots, T(\mathbf{v}_n)$?

Result

Let $V(\mathbb{F})$ and $W(\mathbb{F})$ be two finite dimensional vector spaces. Then V is isomorphic to W iff $dim(V) = dim(W)$.

Example

The vector spaces $\mathbb{R}^n$ and $\mathbb{R}_n[x]$ are not isomorphic.

Let $T : V \to W$ be an isomorphism. Then T is a bijection. Consider $T^{-1} : W \to V$. Thus, for $\mathbf{w} \in W$

$$T^{-1}(\mathbf{w}) = \mathbf{v} \in V \iff T(\mathbf{v}) = \mathbf{w}.$$

Is T^{-1} a linear transformation?

Result

- *If $T : V \to W$ is an isomorphism, then $T^{-1} : W \to V$ is an isomorphism.*
- *If $V \cong W$, then $W \cong V$. We say then V and W are isomorphic.*

Take home

Exercise

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ *and* $S : \mathbb{R}^2 \to \mathbb{R}^2$ *be defined by*

$$T[x, y]^t = [x - y, -3x + 4y]^t \quad \text{and} \quad S[x, y]^t = [4x + y, 3x + y]^t$$

for all $[x, y]^t \in \mathbb{R}^2$. *Show that* S *and* T *are isomorphisms and that* S *is the inverse of* T.

Exercise

Show that

- $T : \mathbb{R} \to \mathbb{R}^2$ *defined by* $T(x) = [x, 0]^t$ *for* $x \in \mathbb{R}$ *is one-one but not onto.*
- $T : \mathbb{R}^2 \to \mathbb{R}$ *defined by* $T[x, y]^t = x$, *for* $[x, y]^t \in \mathbb{R}^2$ *is onto but not one-one.*
- $T : \mathbb{R}^2 \to \mathbb{R}^2$ *defined by* $T[x, y]^t = [-x, -y]^t$, *for* $[x, y]^t \in \mathbb{R}^2$ *is one-one and onto.*

Composition of Linear Transformations

Let $T : U \to V$ and $S : V \to W$ be two linear transformations. The composition of S with T is the mapping $S \circ T : U \to W$ defined by

$$(S \circ T)(\mathbf{u}) = S(T(\mathbf{u})) \quad \text{for all } \mathbf{u} \in U.$$

Result

Let $T : U \to V$ and $S : V \to W$ be two linear transformations. Then the composition $S \circ T$ is also a linear transformation.

The Matrix of a Linear Transformation

Result

Let V and W be two vector spaces with ordered bases

$$B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\} \text{ and } C = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}, \text{ respectively.}$$

Let $T : V \to W$ be a linear transformation, and let

$$A = \Big[[T(\mathbf{v}_1)]_C, [T(\mathbf{v}_2)]_C, \ldots, [T(\mathbf{v}_n)]_C\Big].$$

The $m \times n$ matrix A satisfies

$$A[\mathbf{v}]_B = [T(\mathbf{v})]_C \qquad \text{for all } \mathbf{v} \in V.$$

If $\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{v}_j$, then $T(\mathbf{v}) = \sum_{j=1}^{n} c_j T(\mathbf{v}_j)$. Therefore,

$$[T(\mathbf{v})]_C = \sum_{j=1}^{n} c_j [T(\mathbf{v}_j)]_C = A[\mathbf{v}]_B.$$

Definition

The above matrix A is called the matrix of T with respect to the ordered bases B and C, and is denoted by $[T]_{C \leftarrow B}$.

The statement $[T]_{C\leftarrow B}[\mathbf{v}]_B = [T(\mathbf{v})]_C$ means that the following scheme commutes:

$$\begin{array}{ccc} \mathbf{v} \in V & \xrightarrow{T} & T(\mathbf{v}) \in W \\ \downarrow & & \downarrow \\ [\mathbf{v}]_B \in \mathbb{F}^n & \xrightarrow{T_A} & [T(\mathbf{v})]_C = A[\mathbf{v}]_B \in \mathbb{F}^m \end{array}.$$

Example

Consider $D : \mathbb{R}_3[x] \to \mathbb{R}_2[x]$ *defined by* $D(p(x)) = p'(x)$. *Consider the basis* $B = \{1, x, x^2, x^3\}$ *of* $\mathbb{R}_3[x]$ *and the basis* $C = \{1, x, x^2\}$ *of* $\mathbb{R}_2[x]$. *Then*

$$[D]_{C\leftarrow B} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

Exercise

Let A be an $m \times n$ matrix and consider $T_A : \mathbb{F}^n \longrightarrow \mathbb{F}^m$ given by $T_A(\mathbf{x}) = A\mathbf{x}$. If B and C are the standard bases of $\mathbb{F}^n$ and $\mathbb{F}^m$, respectively, then what is $[T_A]_{C\leftarrow B}$?

Example

If B and C are bases of a finite dimensional vector space V, then the change of basis matrix from B to C satisfies $P_{C\leftarrow B}[v]_B = [v]_C$. Thus, $P_{C\leftarrow B} = [\,I\,]_{C\leftarrow B}$.

Exercise (Take home)

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by

$$T([x, y, z]^t) = [x - 2y, x + y - 3z]^t \quad \text{for } [x, y, z]^t \in \mathbb{R}^3.$$

Let $B = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ and $C = \{\mathbf{e}_2, \mathbf{e}_1\}$ be ordered bases for $\mathbb{R}^3$ and $\mathbb{R}^2$, respectively. Find $[T]_{C\leftarrow B}$ and verify the previous result for $\mathbf{v} = [1, 3, -2]^t$.

Notation: If $V = W$ and $B = C$, then $[T]_{C \leftarrow B}$ is written as $[T]_B$.

Result

Let U, V and W be three vector spaces with ordered bases B, C and D, respectively. Let $T : U \to V$ and $S : V \to W$ be linear transformations. Then

$$[S \circ T]_{D \leftarrow B} = [S]_{D \leftarrow C}[T]_{C \leftarrow B}.$$

Result

Let $T : V \to W$ be a linear transformation between two n-dimensional vector spaces V and W with ordered bases B and C, respectively. Then T is invertible if and only if the matrix $[T]_{C \leftarrow B}$ is invertible. In this case,

$$([T]_{C \leftarrow B})^{-1} = [T^{-1}]_{B \leftarrow C}.$$

Example

Let $T : \mathbb{R}^2 \to \mathbb{R}_1[x]$ be defined by $T([a, b]^t) = a + (a + b)x$ for $[a, b]^t \in \mathbb{R}^2$. Show that T is invertible, and hence find T^{-1}.